DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Lear

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ABSTRACT

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero is trained via large-scale reinforcement learning (RL) directly on base models without preliminary supervised fine-tuning (SFT). We demonstrate that pure RL can incentivize sophisticated reasoning behaviors, including chain-of-thought verification, reflection, self-correction, and generating long thinking tokens before final answers. DeepSeek-R1-Zero achieves 71.0% pass@1 on AIME 2024 and 86.6% on MATH-500. To resolve language mixing and readability issues in pure RL, we develop DeepSeek-R1, which combines cold-start reasoning data with multi-stage RL. DeepSeek-R1 matches OpenAI-o1 across mathematics, coding, and logical reasoning benchmarks. Furthermore, we distill reasoning capabilities into lightweight 1.5B, 7B, 14B, and 32B models, outperforming open-source baselines.

1. Introduction: The Emergence of Pure RL Reasoning

Traditional post-training heavily depends on supervised fine-tuning (SFT) over curated human demonstrations. We investigate whether reasoning capabilities can emerge spontaneously through pure reinforcement learning without human demonstration data. Using Group Relative Policy Optimization (GRPO), we train base language models using rule-based reward signals on mathematical and code benchmarks. Without any SFT examples, DeepSeek-R1-Zero spontaneously learns to generate extensive thinking traces, reconsider initial assumptions, and execute runtime self-correction.

2. Group Relative Policy Optimization (GRPO) Algorithm

Instead of training a critic network of equal size to the policy (as in PPO), GRPO samples a group of G candidate outputs {o_1, o_2, ..., o_G} for query q from the old policy pi_theta_old. The reward r_i is computed using rule-based verifiers (compiler execution correctness for code, numerical equivalence for math). The advantage A_i is normalized across the group: A_i = (r_i - mean(r)) / std(r). This saves enormous memory and eliminates the value network entirely.

3. The DeepSeek-R1 Multi-Stage Training Pipeline

To address language inconsistency and unstructured output in R1-Zero, DeepSeek-R1 adopts a four-stage pipeline:
Stage 1: Cold-start SFT with several thousand high-quality human-readable long chain-of-thought examples.
Stage 2: Reasoning-oriented RL using GRPO on math, logic, and coding benchmarks.
Stage 3: Rejection sampling to generate 800k synthetic reasoning and general-domain examples for comprehensive SFT.
Stage 4: Secondary RL with combined rule-based accuracy rewards and human preference alignment rewards.

3.1 Knowledge Distillation into Dense Architectures

We distill reasoning tokens from DeepSeek-R1 into smaller dense Qwen and Llama architectures (1.5B, 7B, 14B, 32B). Distillation yields superior results compared to training small models with RL from scratch.

4. Benchmark Performance & Mathematical Breakthroughs

DeepSeek-R1 records 79.8% pass@1 on AIME 2024 (matching OpenAI-o1-1217), 97.3% on MATH-500, and 2029 Elo on Codeforces (top 96.3th percentile). The distilled 32B model attains 72.6% on AIME 2024, surpassing all open-source models of comparable scale.

5. Limitations and Reasoning Vulnerabilities

DeepSeek-R1 possesses clear limitations: First, inference latency is high due to long thinking traces (often several thousand reasoning tokens per query). Second, while strong in verifiable STEM domains with rule-based ground-truth rewards, applying GRPO to creative writing or subjective policy tasks remains challenging due to lack of deterministic reward verifiers. Third, overthinking on simple factual queries can lead to unnecessary computational overhead.

6. Future Research Trajectories

Key frontiers include dynamic thinking token budget allocation, extending pure RL reasoning to multi-modal visual proofs, and eliminating reasoning token verbosity through length penalties.